SmartCart Customer Segmentation System

Unsupervised Machine Learning Project Report

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Data Analytics & Machine Learning Project

Executive Summary

SmartCart is a growing e-commerce platform that struggled with generic marketing strategies and inefficient customer targeting. This project developed an Intelligent Customer Segmentation System using Unsupervised Machine Learning to group customers into meaningful segments and support personalized marketing initiatives.

Business Problem

SmartCart lacked visibility into customer behavior patterns, resulting in ineffective campaigns, missed retention opportunities, and difficulty identifying high-value customers.

Project Objectives

• Discover hidden customer behavior patterns
• Segment customers into meaningful groups
• Enable personalized marketing
• Improve customer retention
• Support data-driven decision-making

Dataset Overview

The dataset contains 2,240 customer records and 22 attributes covering demographics, spending behavior, website engagement, campaign response, and customer activity metrics.

Methodology

1. Data Cleaning & Preprocessing
2. Exploratory Data Analysis (EDA)
3. Feature Scaling using StandardScaler
4. PCA for Dimensionality Reduction
5. K-Means Clustering
6. Agglomerative Clustering
7. Elbow Method for optimal cluster determination

Model Development

The Elbow Method identified K=4 as the optimal number of customer segments. Both K-Means and Agglomerative Clustering were evaluated, with Agglomerative Clustering producing better cluster separation and more meaningful customer groups.

Results & Business Insights

• Successfully segmented 2,240 customers into 4 customer personas.

• Cluster 0 (Family Shoppers): Lower spending and weaker campaign response. Recommended: Family bundle offers and discount coupons.

• Cluster 1 (Loyal Premium Customers): High income and high spending. Recommended: Premium memberships and loyalty rewards.

• Cluster 2 (Digital Browsers): High website visits but lower conversions. Recommended: Retargeting campaigns and personalized recommendations.

• Cluster 3 (High ROI Customers): Strong campaign response and purchasing behavior. Recommended: VIP services and personalized engagement.

• Customer segmentation enables personalized marketing, improved retention, optimized marketing spend, and higher ROI.

Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PCA, K-Means Clustering, Agglomerative Clustering, Jupyter Notebook

Conclusion

This project demonstrates how Unsupervised Machine Learning can transform customer data into actionable business insights. The resulting customer personas help SmartCart improve marketing effectiveness, customer retention, and business performance through data-driven decision-making.